# MUHAMMAD ALI HAMZA

Computer Engineering Student

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**LinkedIn:** linkedin.com/in/muhammad-ali-hamza-083b913a5
**GitHub:** github.com/muhammadalihamza2710-sys

## Professional Summary

Computer Engineering student with a strong interest in software engineering, embedded systems, and hardware–software integration. A quick learner with solid analytical and problem-solving abilities, drawn to system-level programming and practical engineering applications. Motivated to build efficient, reliable, and scalable technical solutions.

## Education

**BS Computer Science, FAST NUCES** — Currently Pursuing

**ICS, The Peace Group of Colleges**

## Technical Skills

| | |
|---|---|
| **Programming** | C, C++, File Handling |
| **Engineering & Simulation** | Proteus, Multisim |
| **Systems & Tools** | Linux, Ubuntu, Arch Linux, Operating System Concepts & Handling |
| **Development & Productivity** | Git, GitHub, LaTeX, Prompt Engineering |

## Projects

**Linux Simulator Terminal** — C++

- Developed a command-line Linux simulator in C++ to demonstrate system-level operations and terminal concepts.
- Implemented CLI commands for managing files and directories across simulated C and D drives.
- Added support for selected Windows-style commands to simulate cross-platform command-line behavior.
- Applied core C++ concepts including file handling, functions, conditional logic, loops, and command parsing.

**Smart Electric Safety System** — Engineering Project

- Designed a safety-oriented electrical monitoring system to detect abnormal current leakage conditions.
- Built a monitoring mechanism that provides visual indication of detected electrical faults.
- Implemented an automatic protection mechanism to disconnect the supply upon detecting a dangerous leakage condition.
- Applied concepts of sensors, electrical protection, circuit design, and real-time fault detection.

**Gas Leakage Detection System** — Engineering Project

- Developed a system to identify potentially hazardous gas concentrations in the surrounding environment.
- Integrated gas-sensing and alert mechanisms to provide timely notification upon leakage detection.
- Applied principles of sensors, electronic circuits, signal detection, and safety-oriented system design.
- Focused on improving response time and reducing risks associated with gas leakage.

## Experience

**Intern, FlyRank AI** — Remote

- Currently completing a remote internship at FlyRank AI, gaining practical exposure to a professional technology environment.

## Certifications

- NEBOSH Certification Course

## Professional Activities

- Member, IEEE Society